DECODELABS DATA ANALYTICS INTERNSHIP

E-COMMERCE BUSINESS INTELLIGENCE

SQL DATA ANALYSIS & BUSINESS PERFORMANCE ASSESSMENT

PROJECT 03 | COMPREHENSIVE ANALYTICAL REPORT

PROJECT DETAIL	INFORMATION
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Specialization	Business Intelligence & Data Analytics
Tools	SQLite, DB Browser for SQLite, SQL
Dataset	E-Commerce Orders
Analysis period	January 2023 to June 2025
Report date	October 2026

Prepared as a documented, reproducible SQL business analysis case study.

01 | Executive Summary

This project presents a comprehensive SQL-based analysis of an e-commerce transactional dataset. It examines 1,200 orders associated with 1,189 unique customers from January 2023 through June 2025. A master script of 22 documented SQL queries was developed and successfully executed in DB Browser for SQLite. The analysis evaluates data integrity, product performance, order statuses, payment methods, time trends, marketing channels, customer behavior, and cancellation and return patterns.

Executive Performance Snapshot

PERFORMANCE INDICATOR	RESULT
Total orders	1,200
Unique customers	1,189
Total recorded order value	1,264,761.96
Products / payment methods / referral sources	7 / 5 / 5
Months analyzed	30
Cancelled / returned orders	250 / 247
Combined cancellation and return rate	41.42%
Affected recorded order value	519,673.91
Affected share of order value	41.09%
Repeat customers / rate	11 / 0.93%
Queries developed	22

All financial figures are recorded order amounts across statuses, not confirmed revenue, refunds, or financial losses.

Major Findings

- Cancelled and returned orders account for 497 of 1,200 transactions (41.42%) and 519,673.91 in recorded order value (41.09%).
- Chair led recorded product order value (195,620.11), closely followed by Printer (195,612.61) and Laptop (192,126.56).
- Only 11 of 1,189 observed customers purchased more than once (0.93%), although incomplete customer history limits retention conclusions.
- Gift Card had the highest observed combined cancellation and return rate (44.35%); Online had the lowest (35.27%).
- June 2024 led monthly recorded order value (68,068.54), while April 2023 had the lowest (27,751.71).
- The analyst-defined High Spend segment accounted for 47.63% of total recorded order value.

Executive priority: improve the visibility of cancellation and return reasons, monitor order outcomes consistently, develop comparable customer cohorts, and investigate payment and product differences using additional contextual information.

02 | Project Introduction

2.1 Business Context

E-commerce businesses generate transactional records that can inform decisions about products, customers, marketing, and operational performance. These records become useful when validated, aggregated, interpreted, and presented as business indicators. SQL supports this process with repeatable queries.

2.2 Problem Statement

The original transaction-level dataset does not directly answer which products generate the largest recorded values, how outcomes differ by payment method, how order activity changes monthly, which customers purchase repeatedly, or which product-payment combinations show elevated cancellation and return rates.

2.3 Objectives

- Validate record counts, uniqueness, missing values, numerical ranges, and total-price consistency.
- Compare recorded order activity across products, payment methods, and statuses.

- Analyze monthly order trends with SQL window functions.
- Assess referral sources, coupons, repeat purchase behavior, and spending segments.
- Describe cancellations and returns and translate results into cautious business recommendations.
- Produce one documented and reproducible master SQL script.

2.4 Scope

The project is descriptive and exploratory. It analyzes January 2023 through June 2025 and does not establish causal drivers, realized revenue, profitability, customer lifetime value, or forecasted outcomes.

03 | Dataset Overview

Primary database: DecodeLabs_Project3_SQL_Analysis.db. Primary table: Orders. Dataset size: 1,200 records and 14 columns.

FIELD	DESCRIPTION
OrderID	Unique order identifier
Date	Order date
CustomerID	Customer identifier
Product	Product ordered
Quantity	Ordered units
UnitPrice	Recorded price per unit
ShippingAddress	Shipping destination
PaymentMethod	Method associated with the transaction
OrderStatus	Recorded fulfillment/order status
TrackingNumber	Shipment tracking reference
ItemsInCart	Recorded cart item count
CouponCode	Optional coupon identifier
ReferralSource	Recorded referral channel
TotalPrice	Total recorded order value

3.1 Core Definitions

- Recorded order value: sum of TotalPrice for a specified set of orders, irrespective of status.
- Affected order: an order whose recorded status is Cancelled or Returned.
- Affected order rate: affected-order count divided by total order count in the relevant group.
- Affected order value: sum of TotalPrice for affected orders, not measured financial loss.
- Repeat customer: CustomerID associated with more than one order in the observation window.
- Customer segments: analyst-defined groups based on summed recorded order value.

04 | Data Quality and Validation

Five SQL queries evaluated whether the essential fields were sufficiently consistent for exploratory analysis.

CHECK	OBSERVED RESULT
Total orders / unique OrderIDs	1,200 / 1,200
Duplicate OrderIDs	0
Missing or blank key fields checked	0
Nonpositive quantities / unit prices / total prices	0 / 0 / 0
Quantity range	1 to 5
Unit price range	11.39 to 699.93
Order total range	11.39 to 3,456.40
Total calculation mismatches	0

4.1 Uniqueness and Completeness

COUNT(*) and COUNT(DISTINCT OrderID) confirmed 1,200 unique order identifiers. Nine key columns, including OrderID, Date, CustomerID, Product, Quantity, UnitPrice, TotalPrice, PaymentMethod, and OrderStatus, showed no missing or blank entries in the implemented tests. Optional coupon absences were not treated as erroneous.

4.2 Numerical Consistency

All inspected quantities and prices were positive. Comparing TotalPrice with Quantity multiplied by UnitPrice, with a small decimal tolerance, revealed no mismatches.

4.3 Data Quality Conclusion

The dataset passed the implemented record-level checks. Independent verification of source systems, addresses, settlement records, fulfillment events, and other untested fields was outside scope.

05 | SQL Methodology and Analytical Techniques

The workflow progressed through five sections: data validation (Queries 01 to 05), business performance (06 to 09), time-based analysis (10 to 12), marketing and customer analysis (13 to 17), and order risk analysis (18 to 22).

TECHNIQUE	USE
COUNT, COUNT(DISTINCT)	Record and unique-customer counts
SUM, AVG, MIN, MAX	Values, averages, ranges
GROUP BY, ORDER BY	Aggregation and rankings
CASE WHEN	Conditional metrics and segments
TRIM, NULLIF, COALESCE	Blank handling and safe division
strftime()	Year-month grouping
WITH (CTEs)	Staged analytical calculations
LAG()	Month-over-month change
RANK()	Monthly value ranking
LIMIT	Highest-value individual orders

The 22 statements were consolidated in DecodeLabs_Project3_Complete_SQL_Analysis.sql and executed against the SQLite database. The final query returned 35 product-payment combinations, and the application displayed “Execution finished without errors.”

06 | Business Performance Analysis

6.1 Product Performance

PRODUCT	ORDERS	UNITS	ORDER VALUE
Chair	178	562	195,620.11
Printer	181	542	195,612.61
Laptop	173	535	192,126.56
Tablet	179	497	186,568.95
Monitor	163	480	175,651.41
Desk	170	508	167,459.93
Phone	156	411	151,722.39

Chair had the highest recorded value, whereas Printer had the most orders. Chair also led ordered units. Several categories were closely grouped by recorded value, suggesting category-level performance should be monitored rather than relying solely on the leading product. The figures include orders across all statuses.

6.2 Order Status Distribution

STATUS	ORDERS	SHARE	ORDER VALUE
Cancelled	250	20.83%	276,396.21
Returned	247	20.58%	243,277.70
Pending	237	19.75%	256,328.15
Shipped	235	19.58%	246,159.58
Delivered	231	19.25%	242,600.32

Cancelled and Returned together constituted 41.42% of orders. This is a signal for operational investigation but does not explain why these statuses occurred.

6.3 Payment Method Performance

METHOD	ORDERS	SHARE	AVG ORDER VALUE
Online	258	21.50%	1,017.22
Cash	246	20.50%	1,056.04
Credit Card	234	19.50%	1,127.55
Debit Card	232	19.33%	1,001.56
Gift Card	230	19.17%	1,070.97

Online was the most frequent method, but Credit Card had the highest average recorded order value. Order frequency and average value describe distinct dimensions of business activity.

6.4 Payment-Related Affected Rates

PAYMENT METHOD	CANCELLED + RETURNED RATE
Gift Card	44.35%
Credit Card	44.02%
Cash	43.09%
Debit Card	40.95%
Online	35.27%

Observed differences may reflect transaction mix or other factors. The results do not establish that the payment method causes cancellation or return.

07 | Time-Based Performance Analysis

Orders span 30 calendar months. SQLite strftime() grouped transactions by month; LAG() compared successive months, and RANK() ordered months by recorded value.

INDICATOR	RESULT
Highest month	June 2024: 68,068.54
Second-highest	May 2023: 63,836.84
Third-highest	January 2023: 56,685.75
Lowest month	April 2023: 27,751.71

7.1 Selected Month-over-Month Changes

MONTH	CHANGE
February 2023	-29.23%
May 2023	+130.03%
June 2024	+143.90%
June 2025	+22.24%

Monthly values fluctuate substantially. Without promotional calendars, inventory and marketing context, these changes should not be attributed to particular campaigns or seasonality. Only six months of 2025 are included, so total values cannot be compared directly with full calendar years.

08 | Marketing and Customer Analysis

8.1 Referral Source Performance

SOURCE	ORDERS	SHARE	ORDER VALUE
Instagram	259	21.58%	275,285.45
Email	250	20.83%	261,808.55
Google	241	20.08%	250,441.48
Facebook	228	19.00%	250,410.90
Referral	222	18.50%	226,815.58

Instagram had the largest recorded order count and value among the named referral sources. Without campaign spending, exposures, or conversion denominators, this is not evidence of superior marketing efficiency or return on investment.

8.2 Coupon Usage

COUPON CATEGORY	ORDERS	SHARE
FREESHIP	313	26.08%
No Coupon	309	25.75%
WINTER15	292	24.33%
SAVE10	286	23.83%

“No Coupon” denotes an order without a recorded coupon and is not necessarily missing or erroneous data. Coupon participation alone does not establish incremental demand or profitability.

8.3 Customer Purchase Frequency

METRIC	RESULT
Unique customers	1,189
One-time customers	1,178
Repeat customers	11
Orders placed by repeat customers	22
Observed repeat-customer rate	0.93%

Most observed CustomerIDs appear once. Because the dataset covers a finite period and customer acquisition timing is unavailable, the observed repeat rate should not be presented as a complete retention measure.

8.4 Customer Spending Comparison

METRIC	ONE-TIME	REPEAT
Customers	1,178	11
Orders	1,178	22
Recorded order value	1,245,226.18	19,535.78
Average value per customer	1,057.07	1,775.98

Repeat customers have a higher observed average value per customer, but the group contains only 11 people. These totals reflect recorded orders, not verified lifetime value.

8.5 Spending Segmentation

Exploratory thresholds: Premium Spend at 3,000 or more; High Spend from 1,500 to below 3,000; Medium Spend from 500 to below 1,500; Low Spend below 500.

SEGMENT	CUSTOMERS	ORDER VALUE	SHARE
High Spend	289	602,379.98	47.63%
Medium Spend	493	451,172.43	35.67%
Premium Spend	34	112,184.68	8.87%
Low Spend	373	99,024.87	7.83%

High Spend contributed the greatest recorded value, while Medium Spend contained the largest customer count. The thresholds were selected for this exercise and are not validated customer-business classifications.

09 | Cancellation and Return Risk Analysis

9.1 Affected Rates by Product

PRODUCT	COMBINED RATE
Monitor	43.56%
Tablet	43.02%
Laptop	42.77%
Chair	41.01%
Printer	40.33%
Phone	39.74%
Desk	39.41%

Affected rates are broadly elevated across categories. Monitor is highest, but further investigation is required before inferring product quality or fulfillment causes.

9.2 Overall Order Risk

INDICATOR	RESULT
Total orders	1,200
Cancelled / returned	250 / 247
Affected orders	497
Affected order rate	41.42%
Total recorded order value	1,264,761.96
Affected recorded value	519,673.91
Affected share of value	41.09%

Affected recorded value is not a measured refund or a confirmed loss. No payment settlement or accounting records were analyzed.

9.3 Highest-Value Affected Orders

ORDER ID	PRODUCT	STATUS	ORDER VALUE
ORD201122	Monitor	Returned	3,390.95
ORD200469	Chair	Cancelled	3,384.90
ORD200328	Tablet	Cancelled	3,370.20
ORD200326	Laptop	Returned	3,352.40

These are selected results from the top-ten query and suggest individual transactions to inspect if supporting fulfillment and refund information becomes available.

9.4 Affected Rates by Quantity

QUANTITY	COMBINED RATE
1	44.31%
2	40.83%
3	43.04%
4	37.45%

QUANTITY	COMBINED RATE
5	41.47%

Single-unit orders have the highest observed rate; four-unit orders the lowest. The figures do not show a consistent monotonic relationship or demonstrate causation.

9.5 Product-Payment Combinations

The final query produced 35 combinations and illustrates why examining multiple business dimensions can uncover patterns masked by aggregate measures.

PRODUCT	METHOD	AFFECTED / TOTAL	RATE
Tablet	Cash	19 / 35	54.29%
Printer	Gift Card	15 / 29	51.72%
Laptop	Gift Card	17 / 33	51.52%
Printer	Debit Card	17 / 33	51.52%
Monitor	Credit Card	17 / 34	50.00%

Printer orders paid Online had a comparatively low observed affected rate of 23.08% (9 of 39). Segment sample sizes are modest, so the percentages are exploratory and should be confirmed in larger datasets.

10 | Consolidated Business Findings

Cancellation and return activity

Evidence: 497 affected orders, or 41.42%, with 519,673.91 in associated recorded value.

Recommended follow-up: Capture structured reasons and examine order lifecycle events.

Balanced leading product categories

Evidence: Chair, Printer, and Laptop each contributed roughly 192,000 to 196,000 of recorded order value.

Recommended follow-up: Monitor categories using completed-order outcomes, costs, and return reasons.

Limited observed repeat purchase

Evidence: 11 repeat customers among 1,189 unique observed customers.

Recommended follow-up: Analyze acquisition cohorts and follow-up purchase windows.

Payment-related differences

Evidence: Gift Card 44.35% affected rate versus Online 35.27%.

Recommended follow-up: Control for product, order value, and customer mix in follow-up analyses.

Variability over time

Evidence: Monthly recorded values ranged from 27,751.71 to 68,068.54.

Recommended follow-up: Build consistent monthly dashboards and investigate contextual changes.

Spending concentration

Evidence: High Spend segment represented 47.63% of recorded value.

Recommended follow-up: Validate segment definitions against completed orders and margins.

11 | Strategic Recommendations and Action Plan

11.1 Improve Order Outcome Visibility

Introduce standardized cancellation and return reason codes, track initiating parties, and link outcomes to delivery and customer service events.

Suggested indicators: Cancellation rate, return rate, affected value by category.

11.2 Develop Customer Cohort Reporting

Identify first observed order date, standardize follow-up windows, and compare repeat purchasing within appropriately eligible cohorts.

Suggested indicators: 30-, 60-, and 90-day repeat purchase rates.

11.3 Investigate High-Affected Segments

Review product-payment combinations with sufficient sample sizes and compare order status histories, customer mix, and transaction characteristics.

Suggested indicators: Affected rates and counts by product-payment group.

11.4 Establish Monthly Monitoring

Create a recurring SQL report or dashboard of counts, recorded order values, statuses, and month-over-month changes.

Suggested indicators: Monthly count, status mix, order value, and change.

11.5 Improve Marketing Measurement

Combine referral information with campaign identifiers, costs, completed-order outcomes, and consistent attribution rules.

Suggested indicators: Acquisition cost, conversion, and completed-order value.

PRIORITY	INITIATIVE	TIMING
High	Reason tracking and recurring order-status monitoring	Near term
Medium	Customer cohorts and segment investigation	Medium term
Medium	Campaign and profitability data integration	Medium term
Ongoing	SQL validation and reporting maintenance	Continuous

These timings are illustrative priorities, not commitments from an operating business.

12 | Analytical Limitations

Recorded order value versus revenue

TotalPrice includes Cancelled, Returned, Pending, Shipped and Delivered statuses. Actual payments, refunds and revenue recognition are unknown.

Missing reason-level data

The dataset identifies cancellations and returns, not their causes.

Incomplete customer histories

Purchases outside the observation period and customers' true acquisition dates are unavailable.

Missing campaign costs

Referral sources cannot be evaluated by conversion rate, acquisition cost or marketing ROI.

Profitability unavailable

Product costs, operating costs, delivery costs and fees are absent.

Small subgroups

Some product-payment combinations have under 40 orders and their percentages can be unstable.

Exploratory thresholds

Customer spending segments use analyst-selected cutoffs that have not been validated.

No causal identification

Group-level differences do not show that a payment type, quantity, or product causes a given outcome.

13 | Project Outcomes and Technical Deliverables

The project produced a SQLite database, a documented 22-query master script, data validation, product and payment analysis, monthly window-function calculations, marketing and customer exploration, multi-dimensional order risk assessment, and screenshot evidence of successful execution.

13.1 SQL Query Inventory

QUERY	ANALYSIS
01	Verify Total Order Records
02	Check Duplicate Order IDs
03	Check Missing Values
04	Validate Prices and Quantities
05	Verify Order Total Calculations
06	Sales Performance by Product
07	Order Status Analysis
08	Payment Method Analysis
09	Cancellation and Return Rates by Payment
10	Monthly Order Value Trends
11	Month-over-Month Order Value Changes
12	Ranking Monthly Performance
13	Referral Source Performance
14	Coupon Usage Analysis
15	Customer Purchase Frequency
16	Customer Spending Analysis
17	Customer Spending Segmentation
18	Cancellation and Return Rates by Product
19	Overall Order Risk Summary
20	Highest-Value Cancelled and Returned Orders
21	Cancellation and Return Rates by Quantity
22	Product and Payment Method Risk Analysis

Demonstrated competencies include validation, relational querying, conditional aggregation, CTEs, window functions, descriptive segmentation, KPI calculation, reproducible scripting, and balanced interpretation of results.

14 | Conclusion

This project demonstrates how SQL can turn transaction records into a structured assessment of e-commerce performance. Across 22 documented queries, it examined 1,200 orders by product, status, payment method, month, referral channel, customer, and affected-order segment.

The clearest signal is the elevated observed cancellation and return share, representing 41.42% of transactions and 41.09% of recorded order value. The analysis also identified limited observed repeat purchasing, variation across product-payment combinations, and meaningful monthly changes.

These results support operational monitoring and targeted further investigation. They do not establish causal explanations, realized revenue, or financial losses. The central project outcome is a repeatable analytical process connecting data validation, business indicators, careful interpretation, and recommended next steps.

Appendix A | Project Files

FILE	PURPOSE
DecodeLabs_Project3_SQL_Analysis.db	SQLite project database
DecodeLabs_Project3_Complete_SQL_Analysis.sql	Complete 22-query master script
DecodeLabs_Project3_Dataset.csv	CSV dataset
Dataset for Data Analytics (2).xlsx	Source Excel dataset
29_Product_Payment_Risk_Analysis.sql	Standalone supplementary analysis
Screenshots/SQL_Master_Script_Execution.png	Successful execution evidence

Appendix B | SQL Execution Verification

Environment: DB Browser for SQLite. Database: DecodeLabs_Project3_SQL_Analysis.db. Script: DecodeLabs_Project3_Complete_SQL_Analysis.sql. The full script was run and DB Browser displayed “Execution finished without errors.” The final Product and Payment Method Risk Analysis query returned 35 rows.

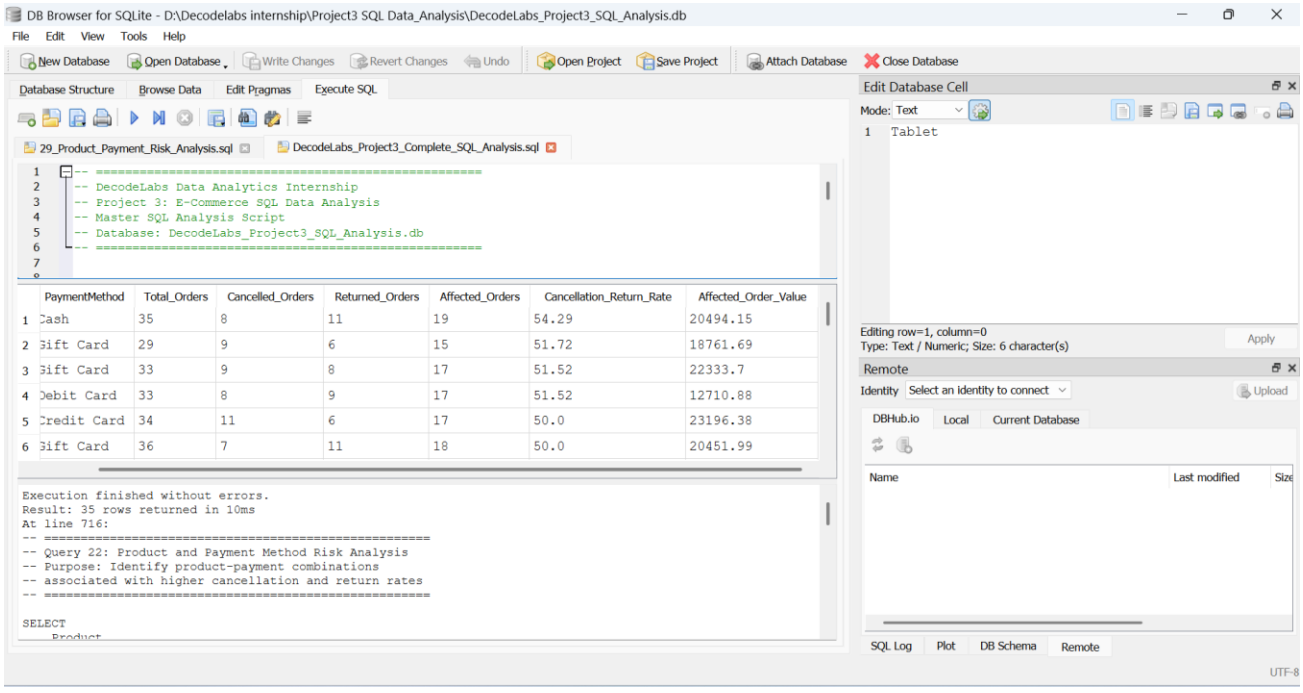


Figure B1. Execution screenshot showing the master script, final query output, and success message.

Appendix C | Future Analytical Extensions

- Order lifecycle timing and fulfillment event tracking.
- Settlement and refund records to distinguish recorded value from financial outcomes.
- Customer acquisition cohorts and controlled repeat-purchase windows.
- Product cost, delivery cost, and margin analysis.
- Campaign cost, traffic, and marketing attribution integration.
- Larger-sample statistical comparisons for product-payment segments.
- Automated reporting and interactive business intelligence dashboards.

END OF REPORT